

Oxidation Process of Bovine Heart Ubiquinol-Cytochrome *c* Reductase as Studied by Stopped-flow Rapid-scan Spectrophotometry and Simulations Based on the Mechanistic Q Cycle Model*

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Stopped-flow rapid-scan spectrophotometry was employed to study complicated oxidation processes of ubiquinol-cytochrome *c* reductase (QCR) that was purified from bovine heart mitochondria and maximally contained 0.36 mol of ubiquinone-10/mol of heme *c*₁. When fully reduced QCR was allowed to react with dioxygen in the presence of cytochrome *c* plus cytochrome *c* oxidase, the oxidation of *b*-type hemes accompanied an initial lag, apparently low potential heme *b*_L was oxidized first, followed by high potential heme *b*_H. Antimycin A inhibited the oxidation of both *b*-type hemes. The oxidation of heme *c*₁ was triphasic and became biphasic in the presence of antimycin A. On the other hand, starting from partially reduced QCR that was poised at a higher redox potential with succinate and succinate-cytochrome *c* reductase, the *b*-type hemes were oxidized immediately without a lag. When the ubiquinone content in QCR was as low as 0.1 mol/mol heme *c*₁ the oxidation of the *b*-type hemes was almost suppressed. As the Q-deficient QCR was supplemented with ubiquinol-2, the rapid oxidation of *b*-type hemes was restored to some extent. These results indicate that a limited amount of ubiquinone-10 found in purified preparations of QCR is obligatory for electron transfer from the *b*-type hemes to iron-sulfur protein (ISP) and heme *c*₁.

The characteristic oxidation profiles of heme *b*_L, heme *b*_H, and heme *c*₁ were simulated successfully based on a mechanistic Q cycle model. According to the simulations the two-electron oxidation of ubiquinol-10 via the ISP and heme *c*₁ pathway, which is more favorable thermodynamically than the bifurcation of electron flow into both ISP and heme *b*_L, does really occur as long as heme *b*_L is in the reduced state and provides ubiquinone-10 at center *i*. Mechanistically this process takes time, thus explaining the initial lag in the oxidation of the *b*-type hemes. With the partially reduced QCR, inherent ubisemiquinone at center *i* immediately oxidizes reduced heme *b*_H thus eliminating the lag.

The mechanistic Q cycle model consists of 56 reaction species, which are interconnected by the reaction paths

specified with microscopic rate constants. The simulations further indicate that the rate constants for electron transfer between the redox centers can be from 10⁵ to 10³ s⁻¹ and are rarely rate-limiting. On the other hand, a shuttle of ubiquinone or ubiquinol between center *o* and center *i* and the oxidation of heme *c*₁ can be rate-limiting. The interplay of the microscopic rate constants determines the actual reaction pathway that is shown schematically by the "reaction map." Most significantly, the simulations support the consecutive oxidation of ubiquinol in center *o* as long as both heme *b*_L and heme *b*_H are in the reduced state. Only when heme *b*_L is oxidized and ISP is reduced can SQ_o donate an electron to heme *b*_L. Thus, we propose that a kinetic control mechanism, or "a kinetic switch," is significant for the bifurcation of electron flow.

Ubiquinol-cytochrome *c* oxidoreductase (QCR¹; *bc*₁ complex or complex III) is one of the three sites for energy conservation in the mitochondrial respiratory chain. This mitochondrial enzyme consisting of 11 subunits (1) translocates protons from the matrix side of mitochondrial inner membranes toward the cytoplasmic side coupled with electron transfer from ubiquinol to cytochrome *c* (2, 3). QCR contains four redox centers, two *b*-type hemes (heme *b*₅₆₂/*b*_H and heme *b*₅₆₆/*b*_L), a *c*-type heme (heme *c*₁), and a 2Fe-2S cluster (ISP) (3). In mitochondria, low potential heme *b*_L is located near the outside surface of the inner membrane, whereas high potential heme *b*_H is situated inside the membrane about 20 Å away from heme *b*_L. ISP and heme *c*₁ are largely extramembranous on the cytoplasmic side of the inner membrane (4). In addition to these four prosthetic groups, ubiquinone-10 is usually found in purified QCR prepared from mammalian mitochondria (5, 6).

Previous work on the mechanism of electron transfer and proton translocation by QCR revealed characteristic features that lead to various hypothetical mechanisms (7, 8). To date, the protonmotive Q cycle, originally proposed by Mitchell (2), has become the generally accepted hypothesis. The central idea of the cyclic model is "branching" of electron flow from ubiquinol that delivers an initial electron to ISP at center *o* on the cytoplasmic side, one of the assumed Q-reactive centers. This electron is transferred from ISP to heme *c*₁ and then to cytochrome *c*. Ubisemiquinone (SQ) thus generated is an intermediate and subsequently donates an electron to low potential heme *b*_L at center *o* yielding quinone in turn. Then, quinone is

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¹ The abbreviations used are: QCR, ubiquinol-cytochrome *c* reductase; CCO, cytochrome *c* oxidase; SCR, succinate-cytochrome *c* reductase; Q_n and Q_nH₂ (*n* = 2, 10), the oxidized and reduced form of ubiquinone-*n*; SQ, semiquinone; ISP, iron-sulfur protein; ET, electron transfer; RS, reaction step.

translocated to center *i*, another Q-reactive site on the matrix side, and reduced to SQ_i by accepting an electron from high potential heme *b*_H to which the electron has been transferred from heme *b*_L (3, 9, 10). Ubiquinone at center *o* or *i* is usually referred to as Q_o or Q_i, respectively. These reactions constitute the first half of the Q cycle. In the second half, the oxidation of another molecule of quinol at center *o* proceeds following the same processes as before and provides a second electron at center *i* to reduce SQ_i to quinol. The quinol then goes back to center *o* completing the protonmotive Q cycle. Thus bifurcation of electron flow from ubiquinol into ISP and heme *b*_L at center *o* as well as electrogenic electron transfer from heme *b*_L to heme *b*_H is most pivotal in the energy conservation mechanism because this supports proton translocation across the membrane.

Despite a considerable experimental support for the protonmotive Q cycle, there are still some observations that are not necessarily compatible with this model (11–15). The oxidation of reduced QCR purified from yeast or bovine heart mitochondria by oxidants has been reported to occur even when the endogenous ubiquinone is depleted from the enzyme complex (11, 12). This result is apparently inconsistent with the proposed functional role of Q not only as a proton translocator but also as an electron mediator between *b*-type hemes and ISP. Also, it has been reported that no lag appears in the oxidation of *b*-type hemes when fully reduced yeast QCR is oxidized by ferricyanide (12), or of heme *b* in rat liver mitochondria by ferricyanide plus air (13). The latter result forms the basis for a proposed semiquinone cycle that assumes a shuttle of ubisemiquinone between center *o* and center *i* (8). According to the Q cycle mechanism, the oxidation of the *b*-type hemes initiated by pulsing an oxidant to the fully reduced QCR does not occur immediately; Q that is required to oxidize heme *b*_H at center *i* is lacking because further oxidation of an intermediate SQ_o by heme *b*_L at center *o* is blocked as this heme is in the reduced state initially. Therefore, it is important to confirm experimentally whether the oxidation of *b*-type hemes accompanies a lag or not and to examine the obligatory role of ubiquinone in this oxidation (16–18).

In previous studies (11–13), nonphysiological oxidants such as ferricyanide and a cobalt chelate were used mostly as an electron acceptor for QCR. Although precautions to eliminate their direct interactions especially with the *b* heme(s) in QCR were taken, the possibility has been pointed out that ferrous heme *b* is accessible to ferricyanide at the outer side of submitochondrial particles (19). This problem would be resolved if we use cytochrome *c* plus cytochrome *c* oxidase (CCO) as the oxidizing system. On the other hand, spectral complexities that are introduced by use of this system and accordingly may hamper the data analyses will be lessened by recording all the spectral changes with rapid-scan spectrophotometry. A series of absorption spectra collected during a reaction and displayed in a way to disclose their characteristic features in detail are useful not only in manifesting the full aspect of the spectral changes at a glance but also in detecting and characterizing reaction intermediates of hitherto unknown nature, if any (20–22). A deliberate choice of experimental conditions to shift the rate-limiting step would be advantageous to increase an occupancy of a certain spectral species and thus to identify its nature.

Here, we employ a stopped-flow and rapid-scan technique to establish whether the oxidation of *b*-type hemes in fully reduced QCR exhibits a lag or not. The oxidation is initiated by mixing reduced QCR with dioxygen in the presence of cytochrome *c* and CCO. The QCR preparations purified from bovine heart muscle usually contain a varying amount of ubiquinone, mostly substoichiometric to heme *c*₁, and consequently the

preparations seem to be a mixture of QCR with a full set of redox centers and an incomplete set. Especially the QCR preparation with the lowest content of ubiquinone is suitable to examine its putative obligatory role in reoxidation, and in fact it is shown that ubiquinone is absolutely necessary for the oxidation of *b*-type hemes. These results have led us to interpret our results in the context of a “Q pocket” model that contains centers *i* and *o*, as assumed by Wikström and Krab (8). Such a pocket may be provided by a dimeric structure of this complex revealed by the structural studies (4, 23) even if it is lacking in the monomeric structure. It is also possible that ubiquinone and ubiquinol make contact alternatively with *o* and *i* domain in the Q pocket without being released from QCR.

We also simulate the behavior of the redox centers based on a proposed kinetic model, in which starting from the fully reduced state of QCR the reaction intermediates proliferate as the reaction proceeds in accordance with the multiplicity of an intermediate containing a specified number of reducing equivalents, and then converge to final products. One of the advantages of this approach is to allow us to test alternative mechanistic possibilities by incorporating the thermodynamic and kinetic parameters that define the reaction into computer simulations. Consequently, a kinetic control mechanism for the bifurcation of electron flow will be proposed that contrasts with the thermodynamic barrier hypothesis proposed by Ding *et al.* (24) but bears some resemblance to the “catalytic switch” mechanism proposed by Brandt *et al.* (25, 26).

EXPERIMENTAL PROCEDURES

Succinate-cytochrome *c* reductase (27), ubiquinol-cytochrome *c* reductase (6), and cytochrome *c* oxidase (20) were purified from beef heart muscle according to previously reported methods and stored at –80 °C until used. Succinate-cytochrome *c* reductase (SCR) contained about 3.5 nmol of heme *b*/mg of protein and exhibited the electron transfer activity of 70–90 nmol of cytochrome *c* reduced·s^{–1}·(mg protein)^{–1} in a medium containing 0.1 M sodium phosphate buffer (pH 7.4), 0.3 mM EDTA, 20 mM succinate, and 100 μM cytochrome *c* (28, 29). Purified preparations of ubiquinol-cytochrome *c* reductase (QCR) contained 4.2–4.5 nmol of heme *c*₁/mg of protein and exhibited the catalytic activity of 330–410 mol of cytochrome *c* reduced·s^{–1}·(mol heme *c*₁)^{–1} when measured in an assay mixture containing 50 mM sodium phosphate buffer (pH 7.0), 0.3 mM EDTA, 50 μM cytochrome *c*, and 25 μM Q₂H₂ at 20 °C (6, 30). The initial rate of cytochrome *c* reduction was corrected against nonenzymatic reduction of cytochrome *c* by ubiquinol in the absence of QCR. The Q₁₀ content of QCR varied with enzyme preparations ranging from 0.1 to 0.36 mol of Q₁₀/mol of heme *c*₁. The purity and the enzyme activity of cytochrome *c* oxidase (CCO) were the same as reported previously (20).

Cytochrome *c* (horse heart, type III) and antimycin A were purchased from Sigma. Cholic acid was obtained from Wako Pure Chemicals (Tokyo, Japan). Other chemicals were obtained commercially at the highest purity.

Stopped-flow rapid-scan recordings with and without a laser flash were carried out on an apparatus as described previously (20–22). With this apparatus the recording time of a spectrum over a 200-nm wavelength range was 12 ms, and usually 100 spectra were recorded with a minimal time interval of 30 ms including the recording time. The dead time of the apparatus under the present experimental conditions was around 10 ms, and the end of this period was chosen as time 0 for displaying spectral changes.

Two different methods were employed to prepare sample solutions for stopped-flow and rapid-scan measurements as follows. Method 1: purified QCR was diluted with 50 mM sodium phosphate buffer (pH 7.4) containing 0.25% sodium cholate to give a final concentration of 6–7 μM heme *c*₁ and placed in one of the reservoirs of the stopped-flow apparatus. The solution was bubbled with N₂ gas for 15 min at 20 °C to attain anaerobic conditions. The fully reduced form of QCR was prepared by addition of a slight excess of sodium dithionite, and allowed to stand for another 15 min at 20 °C under N₂ atmosphere. Then the temperature was lowered to 10 °C. The other reservoir contained cytochrome *c* (1.0 μM) and CCO (6–7 μM aa₃) in an air-saturated solution consisting of 50 mM sodium phosphate buffer (pH 7.4) and 0.25% sodium cholate. After incubation for 10 min at 10 °C, the solutions in both reservoirs were

mixed at a 1:1 ratio to start the reaction, and spectral changes were recorded by rapid-scan spectrophotometry at an appropriate time interval. Method 2: a mixture containing QCR (6–7 μM cytochrome c_1), 1 μM cytochrome c , and CCO (6–7 μM aa_3) in 0.25% sodium cholate, 50 mM sodium phosphate buffer (pH 7.4) was placed in one reservoir and bubbled with a gas mixture of CO and N_2 (1:4) for 10 min at 20 °C. A slight excess of sodium dithionite was added to reduce all redox components before further incubation for 15 min at 20 °C. Then the temperature was brought to 10 °C. In the other reservoir, an air-saturated solution containing 50 mM sodium phosphate buffer (pH 7.4), 0.25% sodium cholate, and no protein components was placed, and the temperature was equilibrated at 10 °C. The two solutions in the reservoirs were mixed to flow at a 1:1 ratio into the observation cell of the stopped-flow apparatus, and the mixture in the cell was irradiated with a laser flash of 587 nm to initiate the redox reaction by photolyzing the cytochrome c oxidase-CO complex. The two methods gave essentially the same result in terms of the oxygen reaction.

Partially reduced QCR was prepared by using a catalytic amount of SCR (0.5 μM cytochrome b) and 10 mM sodium succinate as a reducing system. After reduction of QCR with succinate for 30 min at 20 °C, sodium malonate was added (12 mM, final) to block electron transfer from succinate to QCR catalyzed by SCR. The reaction of partially reduced QCR with cytochrome c and CCO was followed by Method 1.

Optical measurements to determine the electron transfer activity of respiratory enzymes and the concentration of cytochrome components were carried out with a Unisoku multi-wavelength spectrophotometer equipped with a thermostatted cell holder. The ubiquinone-10 content of purified QCR was estimated according to the reported method (31) by using an extinction coefficient of $12.25 \text{ mm}^{-1} \cdot \text{cm}^{-1}$ for the difference between the oxidized and reduced quinone at 275 nm (12). Cytochrome c (6), cytochrome c_1 (32), cytochrome b in SCR (33), and heme a in CCO (20) were determined according to the reported methods. The concentration of CCO was expressed in terms of cytochrome aa_3 . Protein was estimated according to Lowry *et al.* (34).

The spectral data collected by rapid-scan spectrophotometry were processed by using either a home-built computer program as described previously (35) or MATLAB (The MathWorks, Inc., Natick, MA). Numerical calculations for solving the differential equations that represent a reaction model were performed by using MATLAB on a desktop computer (Dell, OptiPlex XMT 5133).

RESULTS AND DISCUSSION

Reaction of Fully Reduced QCR with Dioxygen in the Presence of Cytochrome *c* Plus CCO—Fig. 1A shows three-dimensional display of spectral changes recorded on mixing dithionite-reduced QCR (0.36 mol of Q_{10} /mol of heme c_1) with an air-saturated solution containing cytochrome c and CCO in the stopped-flow rapid-scan apparatus. The concentration of cytochrome c was 6–7-fold lower than that of CCO or of QCR to minimize its spectral contribution to the absorbance change of heme c_1 around 553 nm and to control the rate of its oxidation. An excess of CCO is expected to keep cytochrome c almost in the oxidized state during the redox reaction. The first spectrum taken immediately after the mixing, or at time 0, shows that QCR is almost in the reduced state with a peak of heme b components at 562 nm and a shoulder of heme c_1 at 553 nm. A broad absorption band around 600 nm is due to the oxidized form of CCO. As the reaction proceeds, the 562-nm peak becomes smaller after a short lag period with a growth of the 604-nm peak. This result indicates that mixing of fully reduced QCR with cytochrome c and CCO under the air brought about the oxidation of hemes b in QCR and electron transfer to CCO. It should be noted that a fairly large portion of heme b still remained in the reduced state at 3 s (Fig. 1, A and B).

Based on the ubiquinone content of the samples employed, we simply assume that maximally 36% QCR in the purified preparation contains a full set of redox centers; that is 1 mol each of Q_{10} -(2), heme b_L -(1), heme b_H -(1), heme c_1 -(1), and ISP-(1). In parentheses the maximal number of reducing equivalent that can be retained is given. Accordingly, fully reduced QCR contains six reducing equivalents. It is proposed that center o accommodates two ubiquinone molecules at domains Q_{os} and Q_{ow} that have a strong and weak interaction with

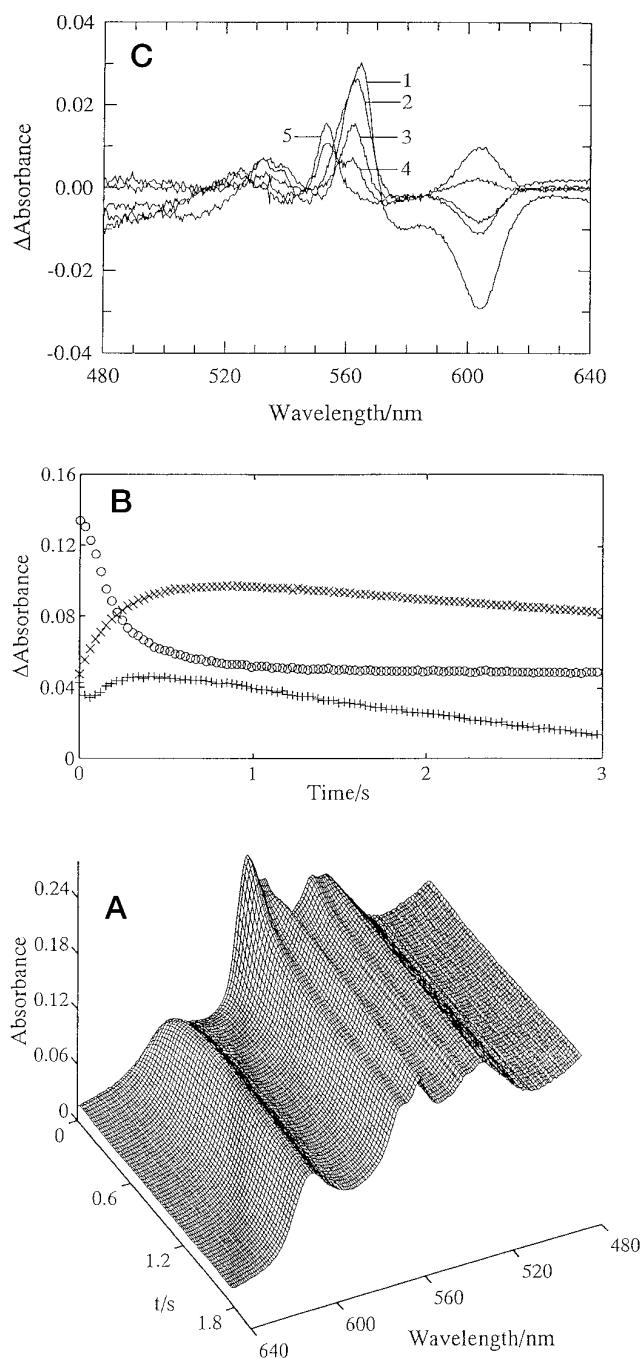


FIG. 1. Oxidation of fully reduced QCR by molecular oxygen in the presence of cytochrome *c* plus CCO. Fully reduced QCR containing 0.36 mol of Q_{10} /mol of cytochrome c_1 was prepared in 0.25% sodium cholate, 50 mM sodium phosphate buffer (pH 7.4) according to Method 1 (see "Experimental Procedures") and mixed with an air-saturated solution containing cytochrome c and CCO. The final concentrations of QCR, cytochrome c , and CCO after mixing were 3.8, 0.5, and 3.0 μM , respectively. Spectral changes were recorded at 10 °C with the time interval of 30 ms. A, a three-dimensional display of the spectral change; B, the absorbance changes at 565–575 nm ($\circ$), 554–540 nm ($+$), and 604–630 nm ($\times$) were taken from the data in A to monitor the redox state of hemes b , heme c_1 , and CCO, respectively; C, absorption spectra in A were processed to give following time difference spectra. 1, 0 – 0.15 s; 2, 0.15 – 0.3 s; 3, 0.3 – 0.6 s; 4, 0.6 – 1.5 s; 5, 1.5 – 3 s.

ubiquinone, respectively (24, 36). The stoichiometry of the present preparation shows that only one of them is retained probably at Q_{os} . On reaction with the oxidizing system, QCR equipped with a full set of redox centers can be oxidized rapidly compared with the incomplete complex. This is in accord with

the experimental result indicating that a fairly large portion of heme *b* remains in the reduced state after the rapid oxidation (Fig. 1, A and B) as described above. According to the Q cycle mechanism ubiquinone is supposed to be released from QCR to move from center *o* to center *i* or vice versa (2, 3). But it is less likely that a limited amount of ubiquinone-10 found in the purified preparation of QCR swims around in a detergent medium to mediate translocation of protons between centers *o* and *i*. Rather, it is more likely for ubiquinone to mediate proton translocation without being released from the QCR complex. Therefore, we postulate in QCR a big ubiquinone pocket that contains centers *i* and *o*, as assumed by Wikström and Krab (8). Ubiquinone and ubiquinol (not ubisemiquinone) may be able to make contact alternatively with *o* and *i* domain in the ubiquinone pocket through a "flip-flop" motion. Such a pocket may be provided by a dimeric structure of this complex as described previously. Further studies are required to find out whether this kind of mechanism is functioning in the catalytic reaction of QCR in phospholipid membranes. Also the x-ray crystallographic studies that are in progress will soon reveal these essential features (4, 37–39).

The time courses of absorbance difference at 566–575 nm, 554–540 nm, and 604–630 nm that monitor the redox change of hemes *b*, heme *c*₁, and CCO, respectively (Fig. 1B), were derived from the data shown in Fig. 1A. Heme *b* components in QCR were oxidized with an initial lag, and a half-time for the oxidation was about 150 ms. Electron transfer to CCO occurred in the same time range as shown by an initial absorbance increase at 604–630 nm. The oxidation of heme *c*₁ was triphasic; the rapid oxidation and re-reduction that occurred within 300 ms after initiation of the reaction was followed by a slow decrease in absorbance for heme *c*₁. A decline of the absorbance of CCO was slower than heme *c*₁. As the oxidation of heme *b* with a lag and the triphasic oxidation of heme *c*₁ were already observed by de Vries *et al.* (40) who employed yeast mitochondrial membranes, the present result confirms their finding.

Fig. 1C shows the time difference spectra derived from the data in Fig. 1A. A difference spectrum between 0 and 150 ms shows a peak at 565 nm and a deep trough at 605 nm (spectrum 1 in Fig. 1C), indicating that heme *b*_L in QCR is oxidized first accompanying electron transfer to CCO. Tsai *et al.* (12) demonstrated, by using a rapid freezing EPR technique, that low potential heme *b*_L was oxidized more rapidly than high potential heme *b*_H during reaction of fully reduced yeast QCR with Q₁. The peak of heme *b* shifts to 563 nm in the difference spectra of 0.15–0.3 and 0.3–0.6 s (spectra 2 and 3), indicating that the oxidation of heme *b*_H dominates after heme *b*_L has been oxidized to an appreciable extent. The oxidation of heme *c*₁ in a later stage is apparent as indicated by the peak at 553 nm in the difference spectra of 0.6–1.5 and 1.5–3 s (spectra 4 and 5).

Fig. 2 shows the effect of antimycin A on the oxidation of QCR (0.36 mol of Q₁₀/mol of heme *c*₁). Fully reduced QCR was prepared in the presence of antimycin A and mixed with an air-saturated solution containing cytochrome *c* plus CCO to initiate the reaction. There was no sign for the oxidation of *b*-type hemes, and the oxidation of heme *c*₁ was biphasic. When the reduction of hemes *b* was incomplete by chance, an oxidant-induced reduction of heme *b*_L occurred although its extent relative to the total heme *b* was less than 5% (Fig. 2). An absorbance increase of CCO at 604 nm due to electron transfer from QCR proceeded along with the oxidation of heme *c*₁, followed by its gradual decay as in Fig. 1.

Oxidation of QCR with a Low Q₁₀ Content—Fig. 3A shows the redox behavior of fully reduced QCR (0.1 mol of Q₁₀/mol of heme *c*₁) upon reaction with oxygen in the presence of cyto-

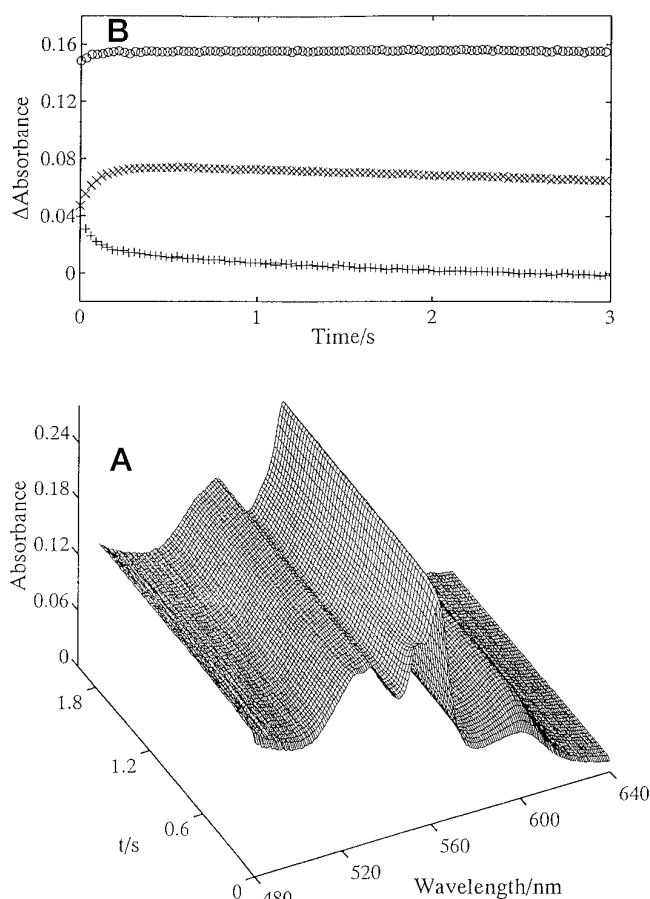


Fig. 2. Effect of antimycin A on electron transfer from fully reduced QCR to CCO. QCR (Q₁₀/cytochrome *c*₁ = 0.36) was reduced by Na₂S₂O₄ in the presence of 10 μ M antimycin A. Other conditions were the same as in Fig. 1. A, a three-dimensional display of the spectral change. B, the absorbance changes at 565–575 nm ($\circ$, hemes *b*), 554–540 nm ($+$, heme *c*₁), and 604–630 nm ($\times$, CCO) were plotted as in Fig. 1B.

chrome *c* and CCO. When the reaction was initiated by the flow-flash method (20, 41), CCO was oxidized completely within a few milliseconds at 10 $^{\circ}$ C (20). Since this oxidation is much faster than that of fully reduced QCR so far examined, no change was seen in QCR during the complete oxidation of CCO. Then, the shoulder at 553 nm disappeared accompanying an increase in absorbance at 604 nm, whereas the peak at 562 nm remained unchanged (Fig. 3A). This result indicates that hemes *b* in QCR with a low Q₁₀ content were not oxidized by the cytochrome *c*/CCO system in the time range examined. However, it should be noted that QCR with the low Q₁₀ content is catalytically active enough to exhibit the electron transfer activity of about 350 nmol of cytochrome *c* reduced $\text{s}^{-1} \cdot (\text{mol heme } c_1)^{-1}$ when assayed in the presence of 25 μ M Q₂H₂ and 50 μ M cytochrome *c* as described under "Experimental Procedures." In accordance with this, when QCR (0.1 mol of Q₁₀/mol of heme *c*₁) was supplemented with exogenous ubiquinone-2 (Q₂) prior to reduction with sodium dithionite, the oxidation of hemes *b* was restored to some extent (Fig. 3B). At the same time, re-reduction of heme *c*₁ was recognized in a later stage. Antimycin A inhibited this oxidation. Therefore, it seems that the exogenous Q₂ fulfills the role of intrinsic Q₁₀ to a certain extent. These results clearly indicate that ubiquinone is an essential component for electron transfer from hemes *b* of QCR to CCO via cytochrome *c*. Previously, we reported that in ascorbate-reduced QCR reconstituted into potassium-loaded phospholipid vesicles, a potassium diffusion potential induced the

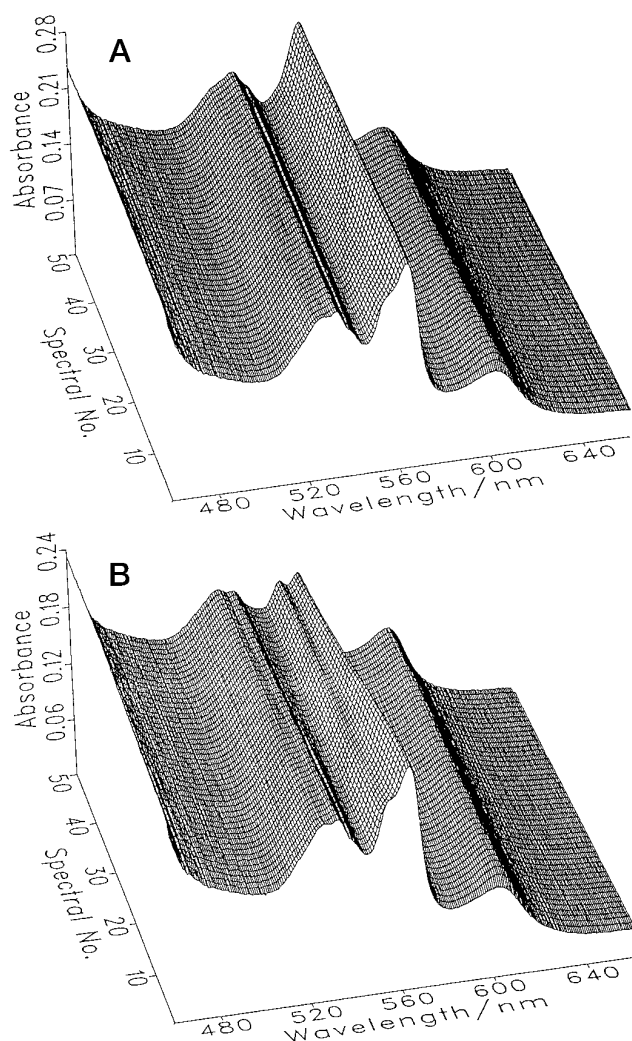


FIG. 3. Oxidation of fully reduced QCR with a lower Q_{10} content by oxygen. A, the aerobic oxidation of $\text{Na}_2\text{S}_2\text{O}_4$ reduced QCR (0.1 mol of Q_{10} /mol of heme c_1) by oxygen in the presence of cytochrome c , and CCO was measured according to Method 2 as described under "Experimental Procedures." The concentrations of QCR, cytochrome c , and CCO after mixing were 3.0, 0.55, 3.0 μM , respectively. Initial 50 spectra are presented at the time interval of 60 ms. B, the enzymes were reduced completely with $\text{Na}_2\text{S}_2\text{O}_4$ in the presence of exogenously added Q_2 ($Q_2/\text{heme } c_1 = 5.0$). Other conditions were as in A.

reduction of heme b components and that ubiquinone was required for reversed electron transfer from heme c_1 to heme b (42). These findings were confirmed recently (43), again supporting this essential role of ubiquinone.

The above conclusion is inconsistent with the previous reports that the oxidation of fully reduced QCR by oxidants was not affected even when the endogenous ubiquinone was depleted from the enzyme (11, 12). It should be noted, however, that either ferricyanide or cobalt chelate was used in these studies. In fact, using the QCR preparation that contained 0.1 mol of Q_{10} /mol of heme c_1 we also observed that both heme b and heme c_1 were oxidized rapidly when the fully reduced form was reacted with 250 eq of ferricyanide (data not shown). This result contrasts with the non-oxidizability of hemes b when cytochrome c plus CCO was the oxidizing system (Fig. 3A) but agrees with the oxidation of these components in yeast QCR that was brought about by 500 eq of ferricyanide (12). Thus, the discrepancy between our results and previous ones (11, 12) would be ascribed to the direct oxidation of heme b components in QCR by nonphysiological oxidants, as pointed out previously (19).

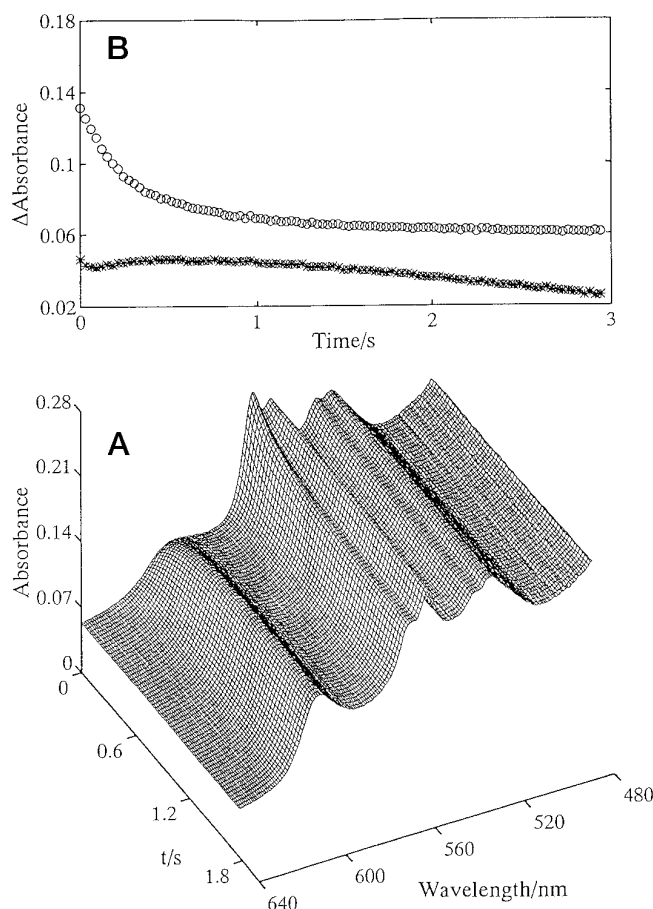


FIG. 4. Reoxidation of partially and fully reduced QCR. A, QCR (6.0 μM heme c_1) containing 0.36 Q_{10} /mol of heme c_1 was reduced partially with succinate in the presence of a catalytic amount of SCR, and the reduction was stopped by addition of sodium malonate as described under "Experimental Procedures." Partially reduced QCR was mixed with an air-saturated solution containing cytochrome c (1.0 μM) and CCO (6.0 μM) to initiate the oxidation. B, the absorbance changes at 562–575 nm ($\circ$, hemes b), and 554–540 nm ($*$, heme c_1) were obtained.

Oxidation of Partially Reduced QCR—Fig. 4A illustrates the redox behavior of partially reduced QCR (0.36 mol of Q_{10} /mol of heme c_1) upon reaction with cytochrome c plus CCO under aerobic conditions. Partially reduced QCR was prepared by incubation of QCR with succinate in the presence of a catalytic amount of succinate-cytochrome c reductase (SCR). Under this condition heme c_1 , iron-sulfur protein, and most of heme b_H were reduced but not heme b_L as reported previously (44). No initial lag was observed during oxidation of heme b_H as shown by a trace at 562–575 nm, and a triphasic redox behavior of heme c_1 was also discerned (Fig. 4B). As a reference, the partially reduced QCR was reduced completely by adding a small amount of dithionite and allowed to react with oxygen catalyzed by cytochrome c and CCO. This time, the oxidation of hemes b clearly accompanied an initial lag that became more apparent compared with the result shown in Fig. 1 (data not shown). The triphasic behavior of heme c_1 was more apparent, too. This oxidation profile was not affected by addition of Q_2 (data not shown). These results indicate that even with the present preparation of QCR (0.36 mol of Q_{10} /mol of heme c_1) the immediate oxidation of heme b occurs as long as the oxidation starts from the partially reduced state.

Summary of Experimental Observations—Stopped-flow rapid-scan spectrophotometric studies on the oxidation kinetics of reduced QCR have revealed characteristic behavior of the re-

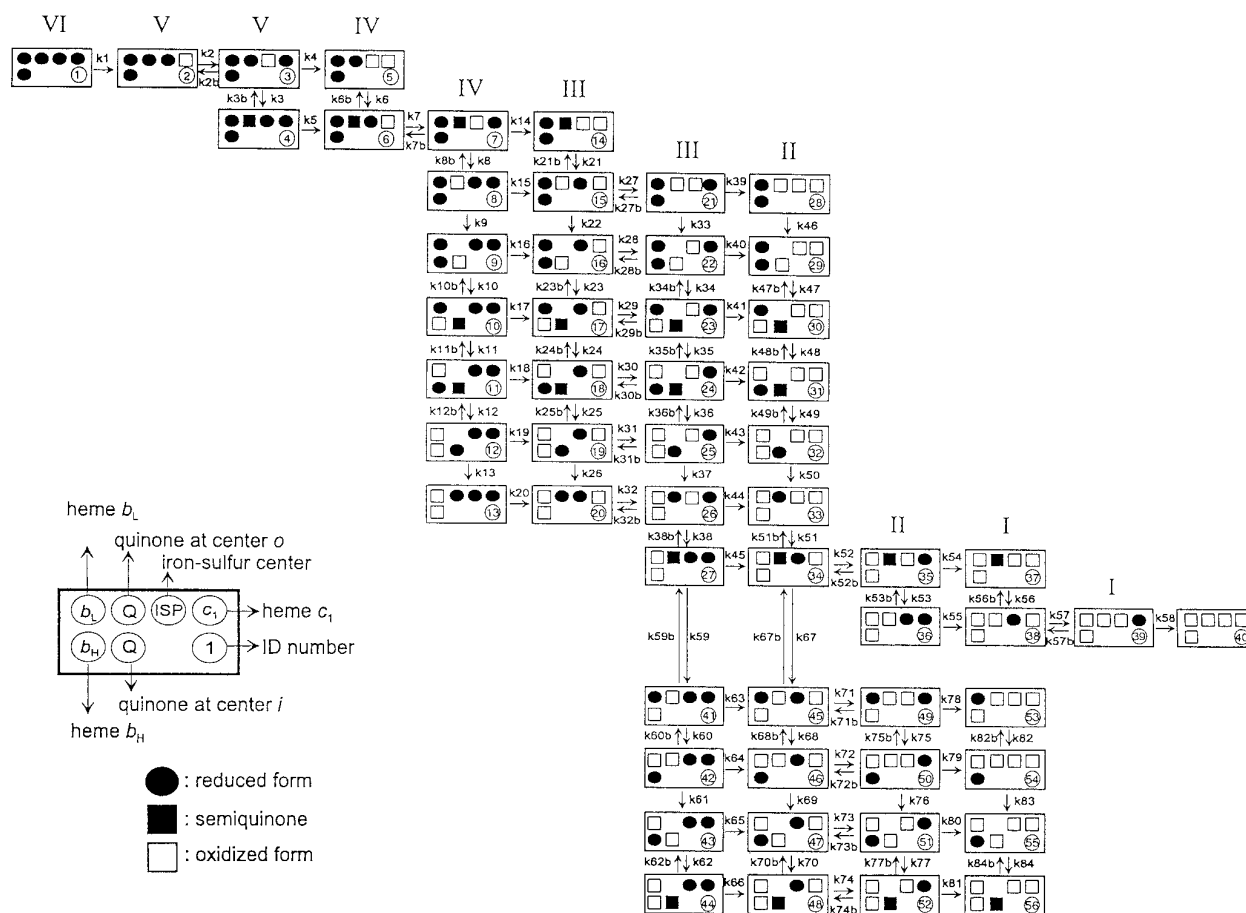


FIG. 5. **The mechanistic Q cycle model for simulating oxidation of QCR.** Each box in this scheme represents a redox state of monomeric QCR that contains heme b_H , heme b_L , heme c_1 , and ISP in addition to ubiquinone-10. The location of every species in the box reflects their physical arrangements in QCR; heme b_L , heme c_1 , and ISP are on the cytoplasmic side of the mitochondrial inner membrane, and heme b_H is inside the membrane; ubiquinone is situated on either center o near heme b_L or center i near heme b_H . Thus, electron transfer is assumed to occur between one redox center and the nearest neighbor.

dox components in the complex as follows. (i) Starting from the fully reduced QCR the oxidation of b -type hemes accompanies an initial lag; apparently heme b_L is oxidized first, followed by heme b_H . Antimycin A inhibits the oxidation. The oxidation of heme c_1 is triphasic and becomes biphasic in the presence of antimycin A. (ii) Q_{10} is essential for electron transfer from b -type hemes to heme c_1 and ISP. (iii) Starting from partially reduced QCR the oxidation of b -type hemes occurs immediately without a lag.

In the sections that follow we try through simulations to explain these observations based on the Q cycle mechanism that we propose in a mechanistic model. It is expected that the redox behavior of heme b_L , heme b_H , and heme c_1 as observed in the present study are reproduced well semiquantitatively, and the behavior of other components would be described too. This model also helps to test the validity of some ideas about the function of QCR and allows predictions.

Reaction Model—Fig. 5 illustrates the reaction scheme for simulating the reaction profiles of hemes b and heme c_1 in a single turnover. Monomeric QCR is assumed to be a functional unit and to behave independently of each other, although the structural studies indicate that QCR is dimeric and suggest some interactions between the monomers (4, 23). The maximum number of different redox states of QCR during the oxidation amounts to 144 if calculated by a simple combination. Out of them 56 states were selected according to the rule that follows and are shown in the model with ID numbers. Each state will be referred to as S1 (State 1), S2, etc. and each reaction step as RS1 (Reaction Step 1), RS2, etc. This selection

rule is based on the Q cycle mechanism in principle that assumes an essential role of ubiquinone in mediating the electron flow from reduced heme b moieties to ISP. Furthermore, we assume some constraints as follows. (i) QCR during the reaction is a closed system that interacts with cytochrome c of the oxidizing system only through heme c_1 . Exchange of ubiquinone with the Q pool is not assumed explicitly. (ii) Either ubiquinol or ubiquinone shuttles (or flip-flops) between center o and center i , and ubisemiquinone generated in either site stays there. (iii) While heme b_L remains in the reduced state ubiquinol gives two electrons consecutively to oxidized ISP that is regenerated by the action of the oxidizing system. (iv) When both heme b_L and ISP are in the oxidized state ubisemiquinone preferentially reduces ISP because this reaction is more favorable thermodynamically. (v) Ubisemiquinone generated in center o reduces oxidized heme b_L , if any, as long as ISP is in the reduced state. (vi) Ubiquinone generated in center o moves to center i without oxidizing reduced heme b_L .

In this reaction scheme (Fig. 5) the fully reduced QCR is at C1R1 if we regard this arrangement as a (sparse) matrix and if the location is represented by a combination of the column (C) and the row (R). The oxidation proceeds toward right and downward. Except for the fully reduced (S1) and fully oxidized (S40) forms, the reaction intermediates having reducing equivalents between 5 and 1 are grouped into two columns for each number of equivalent as shown by a roman numeral at the top of the column. In the left column in each pair every species has oxidized heme c_1 whereas that in the right has reduced heme c_1 ready to be oxidized. Forty members in 10 rows from R1 to R10

TABLE I
 Rate constants used in simulations for the oxidation of reduced QCR

<i>k</i>	<i>k</i> /s ⁻¹	Microscopic rate constants
<i>k</i> _{ox} ^a	40	<i>k</i> ₁ ; <i>k</i> ₁₄ , <i>k</i> ₁₅ , <i>k</i> ₁₆ , <i>k</i> ₁₇ , <i>k</i> ₁₈ , <i>k</i> ₁₉ , <i>k</i> ₂₀ ; <i>k</i> ₅₄ , <i>k</i> ₅₅ ; <i>k</i> ₅₈ ; <i>k</i> ₇₈ , <i>k</i> ₇₉ , <i>k</i> ₈₀ , <i>k</i> ₈₁ <i>k</i> ₄ = <i>k</i> ₅ = <i>k</i> _{ox} /2 <i>k</i> ₃₉ = <i>k</i> ₄₀ = <i>k</i> ₄₁ = <i>k</i> ₄₂ = <i>k</i> ₄₃ = <i>k</i> ₄₄ = <i>k</i> ₄₅ = <i>k</i> _{ox} /20 <i>k</i> ₆₃ = <i>k</i> ₆₄ = <i>k</i> ₆₅ = <i>k</i> ₆₆ = <i>k</i> _{ox} /70 <i>k</i> ₂ , <i>k</i> ₇ ; <i>k</i> ₂₇ , <i>k</i> ₂₈ , <i>k</i> ₂₉ , <i>k</i> ₃₀ , <i>k</i> ₃₁ , <i>k</i> ₃₂ ; <i>k</i> ₅₂ , <i>k</i> ₅₇ ; <i>k</i> ₇₁ , <i>k</i> ₇₂ , <i>k</i> ₇₃ , <i>k</i> ₇₄ <i>k</i> _{2b} , <i>k</i> _{7b} ; <i>k</i> _{27b} , <i>k</i> _{28b} , <i>k</i> _{29b} , <i>k</i> _{30b} , <i>k</i> _{31b} , <i>k</i> _{32b} ; <i>k</i> _{52b} , <i>k</i> _{57b} ; <i>k</i> _{71b} , <i>k</i> _{72b} , <i>k</i> _{73b} , <i>k</i> _{74b} <i>k</i> ₃ , <i>k</i> ₆ , <i>k</i> ₃₈ , <i>k</i> ₅₁ <i>k</i> _{3b} , <i>k</i> _{6b} , <i>k</i> _{38b} , <i>k</i> _{51b} <i>k</i> ₈ , <i>k</i> ₂₁ , <i>k</i> ₅₃ , <i>k</i> ₅₆ <i>k</i> _{8b} , <i>k</i> _{21b} , <i>k</i> _{53b} , <i>k</i> _{56b} <i>k</i> ₉ , <i>k</i> ₂₂ , <i>k</i> ₃₃ , <i>k</i> ₄₆ ; <i>k</i> ₆₁ , <i>k</i> ₆₉ , <i>k</i> ₇₆ , <i>k</i> ₈₃ <i>k</i> ₁₀ , <i>k</i> ₂₃ , <i>k</i> ₃₄ , <i>k</i> ₄₇ ; <i>k</i> ₆₂ , <i>k</i> ₇₀ , <i>k</i> ₇₇ , <i>k</i> ₈₄ <i>k</i> _{10b} , <i>k</i> _{23b} , <i>k</i> _{34b} , <i>k</i> _{47b} ; <i>k</i> _{62b} , <i>k</i> _{70b} , <i>k</i> _{77b} , <i>k</i> _{84b} <i>k</i> ₁₁ , <i>k</i> ₂₄ , <i>k</i> ₃₅ , <i>k</i> ₄₈ ^a ; <i>k</i> ₆₀ , <i>k</i> ₆₈ , <i>k</i> ₇₅ , <i>k</i> ₈₂ <i>k</i> _{11b} , <i>k</i> _{24b} , <i>k</i> _{35b} , <i>k</i> _{48b} ^a ; <i>k</i> _{60b} , <i>k</i> _{68b} , <i>k</i> _{75b} , <i>k</i> _{82b} <i>k</i> ₁₂ , <i>k</i> ₂₅ , <i>k</i> ₃₆ , <i>k</i> ₄₉ <i>k</i> _{12b} , <i>k</i> _{25b} , <i>k</i> _{36b} , <i>k</i> _{49b} <i>k</i> ₁₃ , <i>k</i> ₂₆ , <i>k</i> ₃₇ , <i>k</i> ₅₀ <i>k</i> ₅₉ , <i>k</i> ₆₇ <i>k</i> _{59b} , <i>k</i> _{67b}
<i>k</i> _{pc} ^b	2 × 10 ⁵	
<i>k</i> _{pcb}	<i>k</i> _{pc} /0.32	
<i>k</i> _{qp} ^b	1 × 10 ⁵	
<i>k</i> _{qpb}	<i>k</i> _{qp}	
<i>k</i> _{sp} ^c	1 × 10 ⁵	
<i>k</i> _{spb}	0	
<i>k</i> _{ot} ^d	300	
<i>k</i> _{hq} ^d	1 × 10 ⁵	
<i>k</i> _{hqb}	<i>k</i> _{hq} /4.64	
<i>k</i> _{lh} ^b	2 × 10 ⁵	
<i>k</i> _{lhb}	<i>k</i> _{lh} /46.4	
<i>k</i> _{hs} ^a	1 × 10 ⁵	
<i>k</i> _{hsb}	<i>k</i> _{hs} /0.46	
<i>k</i> _{rt}	50	
<i>k</i> _{sl} ^b	2 × 10 ⁵	
<i>k</i> _{slb}	<i>k</i> _{sl} /68.1	

^a In simulating the oxidation of partially reduced QCR the following assignments were made: *k*_{hs} = 2 s⁻¹, *k*_{ox} = 10 s⁻¹, *k*₁₁ = *k*₂₄ = *k*₃₅ = *k*₄₈ = 0, and *k*_{11b} = *k*_{24b} = *k*_{35b} = *k*_{48b} = 0.

^b The redox potentials used in the calculation of the backward rate constants are as follows: *E*_m(heme *b*_H) = 0.05 V, *E*_m(heme *b*_L) = -0.05 V, *E*_m(heme *c*₁) = 0.25 V, *E*_m(ISP) = 0.28 V (50). *E*_m(Q/QH₂) was assumed to be 0.06 V.

^c If we assume that *k*_{sp} has a definite value, *k*_{spb} can be regarded as nearly zero.

^d In simulating the effect of antimycin A, *k*_{hq} was assumed to be zero.

are grouped into the upper block and the others into the lower block, and these two blocks are connected by the reaction of electron exchange between ubisemiquinone and oxidized heme *b*_L in center *o*. In the upper block S40 is in the fully oxidized state, whereas the most oxidized states in the lower block from S53 to S56 still retain one reducing equivalent either on heme *b* or SQ, although actually this will be lost during a long standing time by a mechanism that cannot be specified.

Simulations of QCR Oxidation—Rate constants used for simulating the oxidation of fully reduced QCR are shown in Table I. (i) The apparent rate constant for the oxidation of heme *c*₁, 40 s⁻¹, was estimated from three exponential analyses of the absorbance change at 554–540 nm in Fig. 1A. (ii) Rate constants for intramolecular electron transfer (ET) among redox centers in QCR were assumed to be in the order of 10⁵ s⁻¹ that belongs to the uppermost estimate in the previous studies (3, 24, 45–48). Simulations indicate, as will be discussed later, that the rate constants for ET can be in the orders lower than 10⁵ s⁻¹ because intramolecular ET is rarely rate-limiting. (iii) The potential differences for a redox pair exchanging an electron were derived from the values in Table I and used to estimate the relative rate constants for intramolecular ET. It is to be noted that no free energy change is assumed for the oxidation of ubiquinol by oxidized ISP ($\Delta E_m = E_{m, SQ/QH} - E_{m, ISP} = 0$) (where *E*_m indicates midpoint redox potential). (iv) A decrease in an apparent rate constant for the oxidation of heme *c*₁ in later stages of the reaction was assumed appropriately to explain both a transient accumulation of reduced heme *c*₁ following the initial rapid oxidation and a slow oxidation that follows (Fig. 1). (v) Translocation of ubiquinone from center *o* to center *i* was equated with an appearance of a species with ubiquinone in center *i* and vice versa. Usually the rate constants for translocation of ubiquinone and ubiquinol, *k*_{ot} and *k*_{rt}, were changed to give the satisfactory result.

Fractional changes of ubiquinol, ubisemiquinone, ubiquinone, reduced ISP, reduced heme *c*₁, reduced heme *b*_H, and reduced heme *b*_L were obtained by summing up every intermediate that contains either of these components (Table II).

Fig. 6 illustrates semiquantitative simulations of three cases, the oxidation of fully reduced QCR in the absence (Fig. 6A) and presence (Fig. 6B) of antimycin A and the oxidation of

partially reduced QCR (Fig. 6C). Fig. 6A reproduces the triphasic oxidation of heme *c*₁, the oxidation of heme *b* with a lag (Fig. 1B), and the oxidation of heme *b*_L prior to heme *b*_H (Fig. 1C) in the absence of antimycin A. The reaction model predicts that the oxidation of heme *b* always accompanies a lag because either ubisemiquinone (SQ_o) or ubiquinone (Q_o) that acts as a direct oxidant of reduced heme *b*_L appears for the first time in the intermediates in R4 (SQ_o) or in R5 (Q_o).

In addition to the redox behavior of heme *b*_H, heme *b*_L, and heme *c*₁, Fig. 6A also simulates a rapid oxidation of ubiquinol with an apparent half-time of 100 ms. This is comparable to the oxidation of fully reduced QCR from yeast by ferricyanide (12). The oxidation profiles of ISP and heme *c*₁ are similar although heme *c*₁ is oxidized faster than ISP. The oxidation of heme *b*_L starts with a lag and almost parallels that of ubiquinol. The oxidation of heme *b*_H lags behind heme *b*_L. This order of oxidation is reverse to the initial oxidation of heme *b*_H followed by heme *b*_L that is assumed in the reaction scheme of Fig. 5. It is highly possible that the immediate filling of the oxidized heme *b*_H with an electron from heme *b*_L by rapid intramolecular ET gives a feature as if heme *b*_L were oxidized first. The reduction level of heme *b*_H stays around 20% for the rest of the reaction. In line with this, the SQ/Q ratio at 3 s after initiation of the reaction is about 4.

The effect of antimycin A on the oxidation of fully reduced QCR is simulated by assuming that the rate constant for the oxidation of reduced heme *b*_H by ubiquinone in center *i* is zero (Fig. 6B). Although the oxidation of heme *c*₁ is simulated as a multiphasic change, it becomes apparently biphasic if the initial rapid changes, which proceed during the dead time of the stopped-flow apparatus, are neglected. The simulation satisfactorily reproduces the non-oxidizability of heme *b*, also. It is reported that inhibition of submitochondrial particles oxidizing succinate by antimycin A induces superoxide anion generation, and the reaction of SQ_o with molecular oxygen is supposed to be the cause (49). This will occur competitively with the consecutive oxidation of SQ_o through the high potential (ISP and heme *c*₁) pathway.

The reaction model also explains the oxidation of partially reduced QCR (Fig. 6C), which is equated to a mixture of S11 and S12. A model for this reaction consists of the intermediates

TABLE II

The reaction intermediates contributing to fractional populations of ubiquinol, ubisemiquinone, ubiquinone, reduced ISP, reduced heme c_1 , reduced heme b_L , and reduced heme b_H

Ubiquinol	Ubisemiquinone	Ubiquinone	ISP	heme c_1	heme b_L	heme b_H
S1, S2, S3, S4	S4, S6, S7, S14	S8, S15, S21, S28	S1, S2; S4, S6	S1, S3; S4, S7	S1, S2, S3, S5	S1, S2, S3, S5
S12, S19, S25, S32	S10, S17, S23, S30	S9, S16, S22, S29	S8, S15; S9, S16	S8, S21; S9, S22	S4, S6, S7, S14	S4, S6, S7, S14
S13, S20, S26, S33	S11, S18, S24, S31	S36, S38, S39, S40	S10, S17; S11, S18	S10, S23; S11, S24	S8, S15, S21, S28	S8, S15, S21, S28
	S27, S34, S35, S37	S41, S45, S49, S53	S12, S19; S13, S20	S12, S25; S13, S26	S9, S16, S22, S29	S9, S16, S22, S29
	S44, S48, S52, S56	S42, S46, S50, S54	S27, S34; S36, S38	S27, S35; S36, S39	S10, S17, S23, S30	S11, S18, S24, S31
		S43, S47, S51, S55	S41, S45; S42, S46	S41, S49; S42, S50	S41, S45, S49, S53	S42, S46, S50, S54
			S43, S47; S44, S48	S43, S51; S44, S52		S43, S47, S51, S55

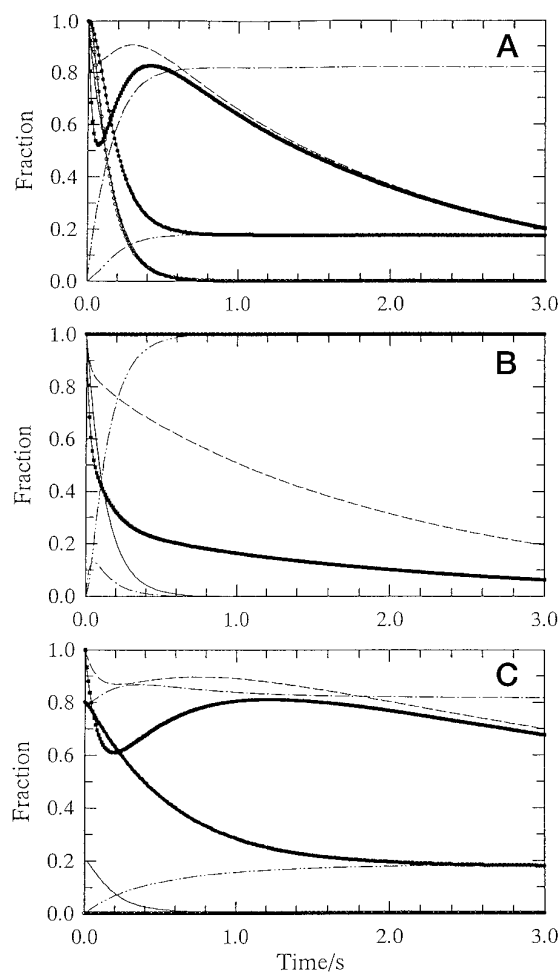


FIG. 6. Simulations for the oxidation of fully reduced QCR in the absence (A) and presence of antimycin A (B), and the oxidation of partially reduced QCR (C). A, rate constants used in the simulation for the oxidation of fully reduced QCR (A) are listed in Table I. B, to mimic the effect of antimycin A k_{hq} was made to zero; C, the oxidation was assumed to start from a mixture of S11 and S12 with the initial condition of S11 = 0.8 and S12 = 0.2. Also $k_{11} = k_{24} = k_{35} = k_{48} = 0$, $k_{11b} = k_{24b} = k_{35b} = k_{48b} = 0$, $k_{hs} = 2 \text{ s}^{-1}$, and $k_{ox} = 10 \text{ s}^{-1}$ were assumed. Fractional changes of ubiquinol (—), ubisemiquinone (---), ubiquinone (· · ·), reduced ISP (— · —), reduced heme c_1 (■), reduced heme b_H (●), and reduced heme b_L (○) were obtained by summing up every reaction intermediate that contains either of these components as listed in Table II.

at and below R6 in the scheme shown in Fig. 5 and was derived by assuming $k_{11} = k_{24} = k_{35} = k_{48} = 0$ and $k_{11b} = k_{24b} = k_{35b} = k_{48b} = 0$. For simulation, the initial condition of S11 = 0.8 and S12 = 0.2 was assumed based on a redox potential of 0.025 V poised by the succinate/SCR system. The rate constant for electron exchange between heme b_H and SQ_i and that for the oxidation of heme c_1 were reduced to $k_{hs} = 2 \text{ s}^{-1}$ and $k_{ox} = 10 \text{ s}^{-1}$ from the values of $1 \times 10^5 \text{ s}^{-1}$ and 40 s^{-1} (listed in Table I), respectively, to comply with slower changes than those in Fig.

1. The simulation shows clearly an immediate oxidation of heme b and a triphasic oxidation of heme c_1 , suggesting that the succinate-fumarate plus SCR system is inhibitory to such processes because of its redox buffering capacity. The successful simulations of the three different experimental results based on the common reaction model encourage us to explore the molecular characteristics of the Q cycle mechanism in more detail as follows.

Mechanistic Characteristics of the Q Cycle Model—The simulation shown by Fig. 6A can be resolved into the rise and decay of each intermediate in the reaction scheme (Fig. 5), and all the reaction intermediates are classified tentatively into six classes according to the maximal fraction that they attain in the time course (class 5 ≥ 0.05 > class 4 ≥ 0.02 > class 3 ≥ 0.01 > class 2 ≥ 0.005 > class 1 ≥ 0.001 > class 0). Among 56 species, 46 intermediates belong to the lower five classes. The most significant 10 intermediates are S1, S2, S3, S4, S11, S18, S43, S44, S55, and S56 and cumulatively can explain most of the simulation result. Their fractional changes as a function of time are shown in Fig. 7A. Reaction intermediates on the upper block (Fig. 7A, top and middle) contribute to the change in the early stage of the oxidation, and those on the lower block (Fig. 7A, bottom) explain most of the later change. In addition to these intermediates, contributions to the reaction by the others are illustrated in Fig. 7B that will be called the reaction map hereafter. The intermediates on the upper block in the reaction scheme are funneled into the lower block through RS59 and RS67. A transient accumulation of S44 is most prominent approaching 0.64 in fraction, and this as well as S43 decays with time being replaced by a growth of S55 and S56 (Fig. 7A, bottom). An accumulation of SQ is mostly represented by S56, suggesting that SQ is stabilized in center i while heme b_H remains in the oxidized state. On the contrary, S55 contains reduced heme b_H . Contributions from more oxidized intermediates on the upper block (S39 and S40) to the spectral change are almost negligible, and the reduction of oxidized heme b_L by SQ in center o that leads to RS59 (S27→S41) and RS67 (S34→S45) must play a crucial role in establishing the reaction pathways.

The simulation results are insensitive to the rate of an elementary ET step as long as the rate constant is between 10^5 and 10^3 s^{-1} , and the ratio between the forward and backward rate constants is kept constant. The simulations are more sensitive to the reaction steps with rate constants around 100 s^{-1} or below that correspond to the oxidation of reduced heme c_1 and translocation of ubiquinone and ubiquinol between center o and center i . The rate of oxidation of heme c_1 was intentionally reduced in the present experiments by decreasing the amount of cytochrome c , an electron mediator between QCR and CCO, and by lowering the temperature to 10°C . Under this specified condition, the translocation rates of ubiquinone and ubiquinol seemingly are most important in controlling the reaction.

Effects of Thermodynamic Barriers on the Oxidation Kinetics—The simulation shown in Fig. 6A and Fig. 7 was done

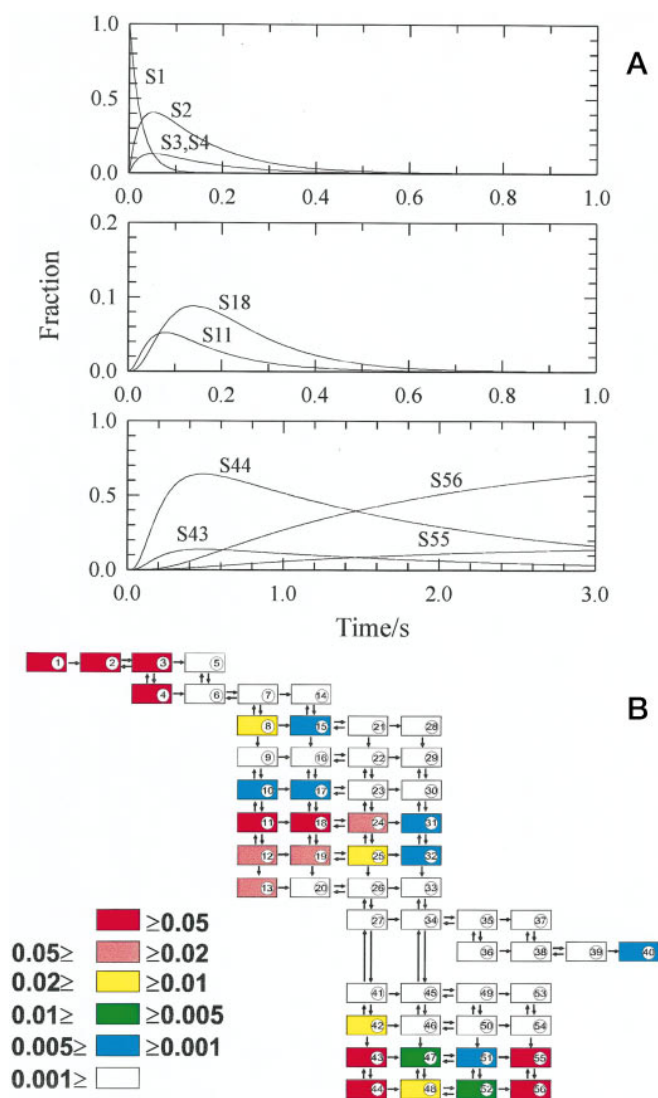


FIG. 7. **Reaction map for the oxidation of fully reduced QCR.** A, time courses for significant reaction intermediates as indicated by ID numbers; B, the reaction map. Each reaction species was classified into six classes according to the maximal fraction that they attain in the time course as follows: class 5 ≥ 0.05 > class 4 ≥ 0.02 > class 3 ≥ 0.01 > class 2 ≥ 0.005 > class 1 ≥ 0.001 > class 0.

successfully by assuming no thermodynamic barrier for the initial oxidation of ubiquinol in center *o* by oxidized ISP. Here a positive thermodynamic barrier is supposed for an endergonic reaction, whereas the negative reaction is for an exergonic process. Ding *et al.* (24), however, assumed that a thermodynamic barrier imposed by the SQ_o formation ($\Delta E_m = E_{m, SQ/QH} - E_{m, ISP} = 0.23$ V) was essential for the bifurcation (24). Thus, the effects of imposing either the positive or negative thermodynamic barrier on the oxidation kinetics were examined based on the present reaction model. When the thermodynamic barrier was $\Delta E_m = 0.1$ V, the oxidation of heme b_L and heme b_H slowed down slightly, and re-reduction of ISP and heme c_1 was also suppressed slightly (Fig. 8A). The others are almost the same as in the case of $\Delta E_m = 0$ (Fig. 6A).

When the imposed barrier was $\Delta E_m = 0.25$ V, which was close to 0.23 V as suggested by Ding *et al.* (24), the oxidation of all redox components was suppressed appreciably (Fig. 8B). Remarkably the oxidation of ubiquinol itself lagged behind that of heme b_L and heme b_H . This is caused mainly by suppression of ET from ubiquinol to ISP, due to a high thermodynamic barrier, even when both ISP and heme c_1 are in the oxidized

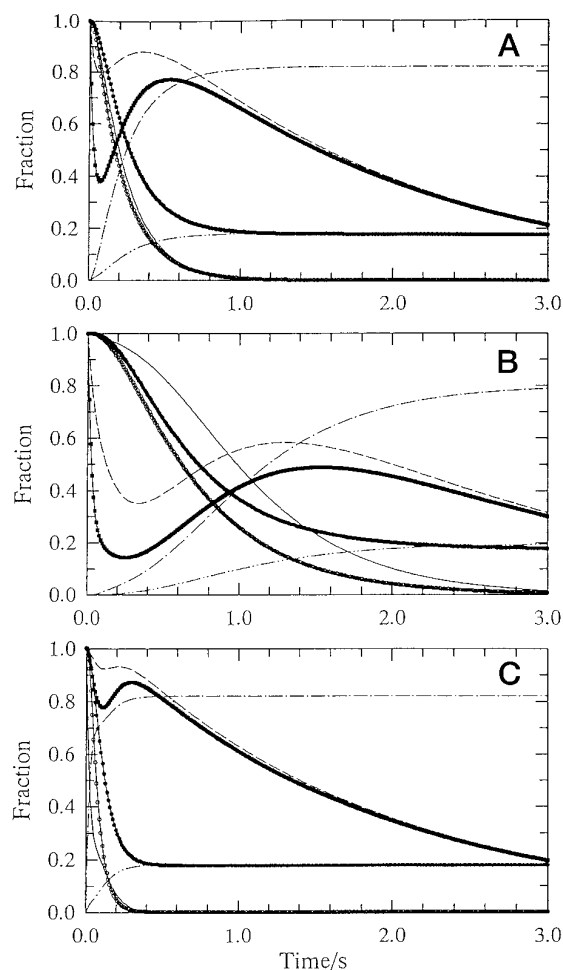


FIG. 8. **Simulation for the oxidation of fully reduced QCR with varying thermodynamic barriers.** A, the positive barrier of $\Delta E_m = 0.1$ V was imposed on the reduction of ISP by ubiquinol in center *o*; B, the positive barrier was $\Delta E_m = 0.25$ V; C, the negative thermodynamic barrier was $\Delta E_m = -0.1$ V. Time courses are as follows for ubiquinone (—), ubisemiquinone (---), ubiquinol (···), reduced ISP (- · -), reduced heme c_1 (■), reduced heme b_H (●), and reduced heme b_L (○).

state. Such a high barrier cannot exist because the simulation deviates appreciably from the experimental result (compare Fig. 8B with Fig. 1).

When the negative thermodynamic barrier was imposed ($\Delta E_m = -0.1$ V), the oxidation of heme b_L and heme b_H were accelerated appreciably. In accordance with this, the re-reduction of ISP and heme c_1 was also promoted (Fig. 8C). Fig. 9 depicts how ubisemiquinone is generated depending on the thermodynamic barrier imposed. SQ_o is generated only once in the initial stage of the reaction, and this is shown by the time course of S4 because contributions from S6, S7, and S14 containing SQ_o are negligible. On the other hand, SQ_i is generated twice as shown by transient formation of S11, S18, S24, and S31 and by S44, S48, S52, and S56, although the extents of contribution are variable. Thus, generation of SQ_o in the initial stage of the reaction is appreciable with $\Delta E_m = -0.1$ V as expected (Fig. 9A). Even when $\Delta E_m = 0$, SQ_o is generated transiently to a maximal fraction of above 0.1 (Fig. 9B). Only with the positive barrier the level of SQ_o becomes negligibly low (Fig. 9C). In this case only SQ_i is detected. Therefore, if the non-occurrence of SQ_o is proved by the ESR measurement (17, 29), it is a sure sign for the existence of the positive thermodynamic barrier although its advantage in controlling the reaction is not clear.

Thermodynamic Barrier Versus Kinetic Control—We have

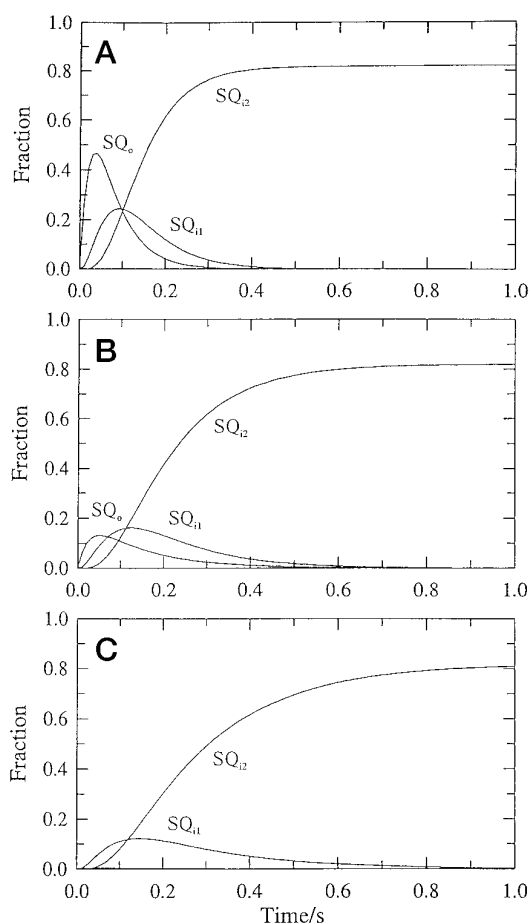


FIG. 9. Dependence of SQ generation on the nature of the thermodynamic barrier. A, $\Delta E_m = -0.1$ V; B, $\Delta E_m = 0$ V; C, $\Delta E_m = 0.1$ V. SQ_{i1} and SQ_{i2} signify SQ that is generated for the first time and second time, respectively, in center *i*.

shown that starting from fully reduced QCR, ubiquinol in center *o* participates in two kinds of reactions (Reactions 1 and 2).



Reaction 1 (RS3, RS6, RS8, and RS21; $\Delta G_0' = -42.5$ kJ/mol) is thermodynamically more favorable than Reaction 2 (RS38, RS51, RS59, and RS67; $\Delta G_0' = -10.6$ kJ/mol), but it is mentioned that this is not observed so far (10). Reaction 1, however, plays a key role in the oxidation of heme b_L and heme b_H in fully reduced QCR because this provides Q_i to oxidize reduced heme b_H in center *i*. The simulation with $\Delta E_m = 0$ as described above indicates that the consecutive oxidation of ubiquinol in center *o* proceeds through oxidized ISP while this is regenerated and both heme b_L and heme b_H are in the reduced state. S8 appears transiently to a maximal extent of 0.0153 in fraction, and this result indicates that the reaction passes through S8 without being trapped. If translocation of ubiquinone from center *o* to center *i* is suppressed by any effective means a transient accumulation of S8 will be increased. Phenomenally, when a negative barrier of $\Delta E_m = -0.1$ V is imposed, the maximal fraction of S8 increases to 0.028. Even with $\Delta E_m = 0.1$ V, the maximal fraction of S8 is 0.0107. As the transient accumulation of S8 is a sure sign for the consecutive oxidation of ubiquinol by Reaction 1, it is concluded that irrespective of the existence of the thermodynamic barrier ubiquinol in center *o* is oxidized consecutively through ISP as long as heme b_L remains in the reduced state.

The production of superoxide anion from submitochondrial particles that oxidize succinate in the presence of antimycin A (49) may be taken as a sign for the incomplete oxidation of ubiquinol and, accordingly, to indicate that further oxidation of SQ_o to Q is suppressed. In fact, the simulation with $\Delta E_m = 0$ indicates that SQ_o is generated transiently to a maximal fraction of 0.13, and this may have a chance to react with molecular oxygen to generate superoxide anion. If SQ_o escapes the attack by oxygen, it will be oxidized to ubiquinone through the high potential (ISP and heme c_1) pathway.

Reaction 2 occurs for S26 and S33 in which both ISP and heme b_L are oxidized with ubiquinol being in center *o*. This is the situation that originated from ubiquinol bifurcation of electron flow into ISP and heme b_L is expected to occur during turnover of QCR. First, starting from S26, RS38 proceeds because this is favorable thermodynamically. Then RS59, ET from SQ_o to oxidized heme b_L , must follow as this is essential to explain the emergence of S43, S44, S55, and S56 on the lower block. It is to be noted that RS59 (and RS67) proceeds only when ISP is in the reduced state, and it is likely that this situation is brought about under a delicate kinetic control. Brandt *et al.* (25, 26) have proposed a catalytic switch mechanism as a prerequisite for the bifurcation of the electron pathway at center *o* and speculated that ISP in the reduced state directs the electron flow into heme b_L stressing its important role in the timing of the reaction. Our analyses essentially support this proposal and furthermore suggest that the bifurcation can be purely a kinetic event. Because of this, we would like to propose a "kinetic switch" mechanism. Whether conformational changes are involved in the switching mechanism as assumed by Brandt *et al.* (25, 26) should be answered by further studies.

A concern of the kinetic control is shown by the following example with the previous case of $\Delta E_m = 0$ as a reference (Figs. 6A and 7B). The rate constant for ET from SQ_o to heme b_L in S27, k_{s1} , was changed to 1 s^{-1} to make this comparable to $k_{45} = 40/20 \text{ s}^{-1}$ which is the rate constant for oxidation of heme c_1 in S27 (Table I). All the other constants were kept unchanged. Although the oxidation profiles of heme b_L and heme b_H were almost the same as in the reference, the oxidation of both ISP and heme c_1 slowed down appreciably. Apparently, the oxidation of heme c_1 is almost biphasic (Fig. 10A). At the same time, the oxidation of ubiquinol was rapid for the initial one-third but stalled temporarily and was followed by a slower oxidation. During the stall, the extent of SQ generation approached a maximal level transiently up to 0.3 in fraction and approached a constant level. Contrary to ubiquinol and SQ, ubiquinone was generated gradually with an initial lag. At 3 s the relative fraction of quinone to ubisemiquinone was about 4 contrary to $\frac{1}{4}$ observed in the reference. On the other hand, during the turnover the transient appearance of S20, S26, and S27 became apparent in the reaction map (Fig. 10B) caused by the kinetic barrier imposed on RS59 and RS67. Consequently, S40, which is a fully oxidized species, was formed to an appreciable extent, and accordingly, an accumulation of S56 became less significant. These clearly indicate the significance of the kinetic control mechanism.

CONCLUSION

The simplified reaction scheme based on the Q cycle mechanism and shown in Fig. 5 explains satisfactorily the following experimental observations. (i) Starting from the fully reduced QCR the oxidation of *b*-type hemes accompanies an initial lag; apparently heme b_L is oxidized first, followed by heme b_H . Antimycin A inhibits the oxidation. The oxidation of heme c_1 is triphasic and becomes biphasic in the presence of antimycin A. (ii) The substoichiometric amount of Q_{10} is essential for elec-

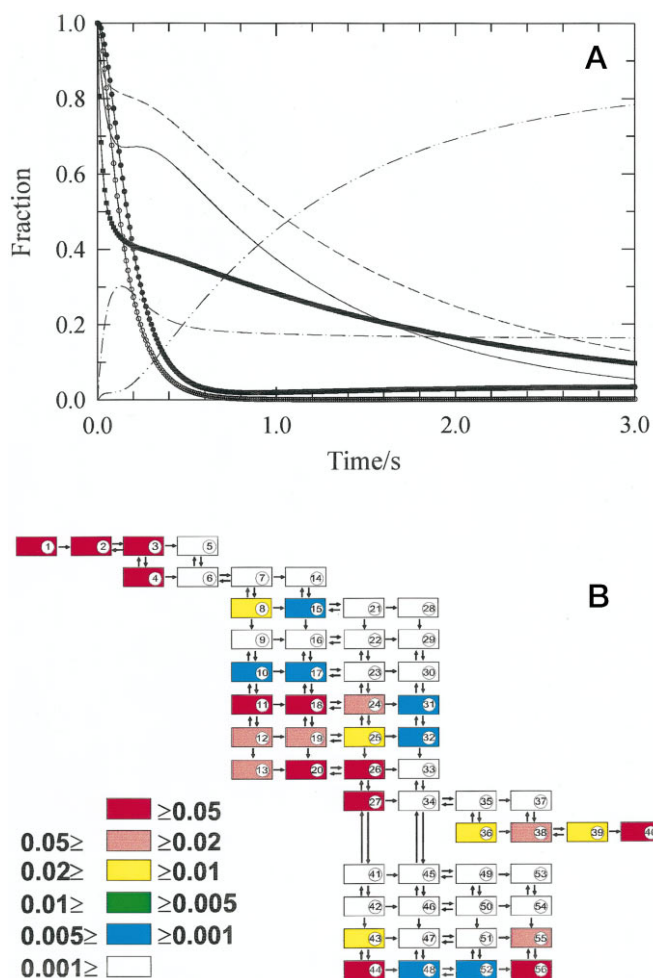


FIG. 10. Simulation for the oxidation of fully reduced QCR. Rate constants used for the simulation were the same as listed in Table I except that k_{sl} , the rate constant for ET from SQ_o to heme b_L in S27, was changed to 1 s^{-1} . A, time courses for ubiquinone (—), ubisemiquinone (---), reduced ISP (— · —), reduced heme c_1 (■), reduced heme b_H (●), and reduced heme b_L (○). B, the reaction map.

tron transfer from b -type hemes to ISP and heme c_1 . (iii) Starting from partially reduced QCR the oxidation of b -type hemes occurs immediately without a lag.

Based on the simulation studies the following mechanisms are proposed. When both heme b_L and heme b_H are in the reduced state ubiquinol in center o is oxidized consecutively by the high potential (ISP and heme c_1) pathway, providing ubiquinone in center i . This initiates the delayed oxidation of heme b_H followed by heme b_L . Even if the consecutive oxidation pathway does function, however, the kinetic control mechanism directs an electron flow preferentially into oxidized heme b_L when this becomes available to SQ_o . The key point of this mechanism is that the fate of SQ_o is determined by the redox state of ISP; only when ISP is in the reduced state SQ_o can act as an electron donor to heme b_L . Thus, ISP is regarded as a redox-linked switch. No endergonic process is required for the bifurcation to occur.

The current mechanistic Q cycle model has proved its use-

fulness in allowing the semiquantitative analysis of the redox behavior of the metal centers and intrinsic ubiquinone in QCR, and in predicting the kinetic mechanism that controls the electron flow. This approach will become a powerful tool in studying the molecular mechanism of energy conservation if elaborated for more rigorous and quantitative treatment of the dynamic data.

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